# Special Relativity Notes

Transformation of the Magnetic Field

9/11/26

**Velocities and symbols used throughout (please read first)**

- $S$ — the original frame; $S'$ — a frame moving along the $+x$ axis with velocity $\vec{v}$ **relative to** $S$ (a boost along $x$). $v = |\vec{v}|$.
- $\gamma = \dfrac{1}{\sqrt{1 - v^2/c^2}}$, $\beta = \dfrac{v}{c}$ ($c$ = speed of light). These always refer to the **frame** velocity $v$.
- $\vec{u}\,' = (u'_x, u'_y, u'_z)$ — velocity of the **test charge** $q$ as measured in $S'$.
- $\vec{u} = (u_x, u_y, u_z)$ — velocity of the same test charge as measured in $S$.
- $\vec{F}$, $\vec{F}\,'$ — force on the test charge in $S$ and in $S'$; $\vec{E}, \vec{B}$ and $\vec{E}\,', \vec{B}\,'$ — the fields in $S$ and in $S'$.
- Unprimed quantities are measured in $S$; primed quantities in $S'$.

## 1 Objective and Tools from the Previous Lecture

**Objective:** to obtain $B'_x$, $B'_y$, $B'_z$ in terms of $(E_x, E_y, E_z)$ and $(B_x, B_y, B_z)$.

The same method as for $\vec{E}\,'$ is used: place a test charge in a chosen state of motion, write the Lorentz force in $S'$ and in $S$, and compare using the force-transformation rule. Three results from the previous notes are needed.

**(a) Lorentz force (in any frame)**

$$\vec{F} = q\left(\vec{E} + \vec{u} \times \vec{B}\right)$$

**where:** $q$ = charge (the same in all frames); $\vec{E}, \vec{B}$ = fields measured in the frame used; $\vec{u}$ = velocity of the charge in that frame.

**(b) Transformation of force ($S \to S'$)**

$$F'_x = \frac{F_x - \dfrac{v}{c^2}\,(\vec{F}\cdot\vec{u})}{1 - \dfrac{v\,u_x}{c^2}}, \qquad F'_y = \frac{F_y}{\gamma\left(1 - \dfrac{v\,u_x}{c^2}\right)}, \qquad F'_z = \frac{F_z}{\gamma\left(1 - \dfrac{v\,u_x}{c^2}\right)}$$

**where:** $\vec{F}\cdot\vec{u} = F_x u_x + F_y u_y + F_z u_z$ is the power delivered to the charge; $u_x$ is the charge's $x$-velocity in $S$; $v/c^2 = \beta/c$.

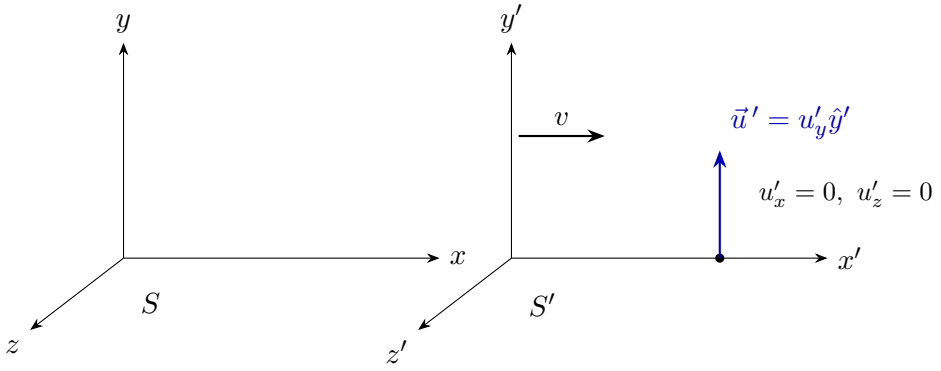
**(c) Velocity of the charge in $S$ from its velocity in $S'$**

$$u_x = \frac{u'_x + v}{1 + \dfrac{v\,u'_x}{c^2}}, \qquad u_y = \frac{u'_y}{\gamma\left(1 + \dfrac{v\,u'_x}{c^2}\right)}, \qquad u_z = \frac{u'_z}{\gamma\left(1 + \dfrac{v\,u'_x}{c^2}\right)}$$

And from the previous lecture (the electric field):

$$E'_x = E_x, \qquad E'_y = \gamma\,(E_y - vB_z), \qquad E'_z = \gamma\,(E_z + vB_y)$$

## 2 Thought Experiment 1: charge moving along $y'$ in $S'$




The test charge moves in $S'$ with $\vec{u}\,' = (0,\, u'_y,\, 0)$.

### Force in $S'$

$$\vec{F}\,' = q\left(\vec{E}\,' + \vec{u}\,' \times \vec{B}\,'\right), \qquad \vec{u}\,' \times \vec{B}\,' = (u'_y B'_z,\ 0,\ -u'_y B'_x)$$

$$\begin{aligned} F'_x &= q\left(E'_x + (\vec{u}\,' \times \vec{B}\,')_x\right) = q\left(E'_x + u'_y B'_z\right) \\ F'_y &= qE'_y \\ F'_z &= q\left(E'_z - u'_y B'_x\right) \end{aligned}$$

### Velocity of the charge in $S$

Apply (c) with $u'_x = 0,\ u'_z = 0$:

$$u_x = v, \qquad u_y = \frac{u'_y}{\gamma}, \qquad u_z = 0$$

**Force in $S$**

With $\vec{u} = (v,\ u'_y/\gamma,\ 0)$, $\vec{u} \times \vec{B} = \left(\frac{u'_y}{\gamma}B_z,\ -vB_z,\ vB_y - \frac{u'_y}{\gamma}B_x\right)$, so

$$F_x = q\left(E_x + \frac{u'_y}{\gamma}B_z\right)$$
$$F_y = q\left(E_y - vB_z\right)$$
$$F_z = q\left(E_z + vB_y - \frac{u'_y}{\gamma}B_x\right)$$

Either set of equations will drive you to the correct result, since the velocity transformation is available. Below, the force-transformation rule (b) is used; the inverse route is checked in Section 5.

**Using the force transformation: the $x$-component gives $B'_z$**

Since $u_x = v$: $1 - \frac{v\,u_x}{c^2} = 1 - \frac{v^2}{c^2} = \frac{1}{\gamma^2}$, and

$$\vec{F} \cdot \vec{u} = F_x\,v + F_y\,\frac{u'_y}{\gamma}$$

Then (b) gives

$$F'_x = \frac{F_x - \frac{v}{c^2}\left(F_x v + F_y\frac{u'_y}{\gamma}\right)}{1 - \frac{v^2}{c^2}} = F_x - \frac{\gamma\,v\,u'_y}{c^2}\,F_y$$

Substitute $F_x = q\left(E_x + \frac{u'_y}{\gamma}B_z\right)$ and $F_y = q(E_y - vB_z)$:

$$F'_x = q\left[E_x + \frac{u'_y}{\gamma}B_z - \frac{\gamma\,v\,u'_y}{c^2}\left(E_y - vB_z\right)\right]$$
$$= q\left[E_x + u'_y B_z\left(\frac{1}{\gamma} + \frac{\gamma v^2}{c^2}\right) - \frac{\gamma\,v\,u'_y}{c^2}E_y\right]$$

Using $\frac{1}{\gamma} = \gamma\left(1 - \frac{v^2}{c^2}\right)$, we get $\frac{1}{\gamma} + \frac{\gamma v^2}{c^2} = \gamma$, so

$$F'_x = q\left[E_x + \gamma\,u'_y\left(B_z - \frac{v\,E_y}{c^2}\right)\right]$$

Compare with $F'_x = q\left(E'_x + u'_y B'_z\right)$ and use $E'_x = E_x$. Comparing the coefficients of $u'_y$:

$$\boxed{B'_z = \gamma\left(B_z - \frac{v\,E_y}{c^2}\right)}$$

**where:** $B_z, E_y$ are measured in $S$; $B'_z$ in $S'$; $v$ is the velocity of $S'$ relative to $S$; $c$ is the

speed of light; $\gamma = 1/\sqrt{1 - v^2/c^2}$.

### Using the force transformation: the $z$-component gives $B'_x$

From (b) with $u_x = v$: $F'_z = \gamma F_z$. Therefore

$$q\left(E'_z - u'_y B'_x\right) = \gamma\, q\left(E_z + vB_y - \frac{u'_y}{\gamma} B_x\right) = q\left[\gamma\left(E_z + vB_y\right) - u'_y B_x\right]$$

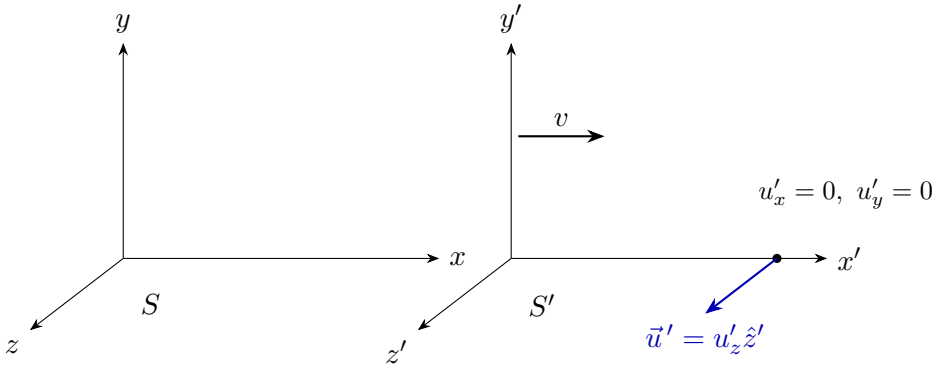
With $E'_z = \gamma(E_z + vB_y)$ the $E$-terms cancel, leaving $-u'_y B'_x = -u'_y B_x$:

$$\boxed{B'_x = B_x}$$

**where:** $B_x$ is the component of $\vec{B}$ along the boost direction; it is the same in $S$ and $S'$.

**Note:** for getting $B'_y$ we need another thought experiment, with $\vec{u}\,' = u'_z \hat{z}'$.

## 3 Thought Experiment 2: charge moving along $z'$ in $S'$




Now $\vec{u}\,' = (0,\, 0,\, u'_z)$. From (c): $\quad u_x = v, \quad u_y = 0, \quad u_z = \dfrac{u'_z}{\gamma}$.

### Force in $S'$

$\vec{u}\,' \times \vec{B}\,' = (-u'_z B'_y,\; u'_z B'_x,\; 0)$, so

$$F'_x = q\left(E'_x - u'_z B'_y\right), \qquad F'_y = q\left(E'_y + u'_z B'_x\right), \qquad F'_z = qE'_z$$

### Force in $S$

$\vec{u} \times \vec{B} = \left(-\dfrac{u'_z}{\gamma} B_y,\; \dfrac{u'_z}{\gamma} B_x - vB_z,\; vB_y\right)$, so

$$F_x = q\left(E_x - \frac{u'_z}{\gamma} B_y\right), \qquad F_y = q\left(E_y - vB_z + \frac{u'_z}{\gamma} B_x\right), \qquad F_z = q\left(E_z + vB_y\right)$$

**Note on signs (differences from the board)**

The board lists $F_z = q(E_z - vB_y)$, $F'_y = q(E'_y - u'_z B'_x)$ and $F_y = q(E_y - vB_z)$ for this experiment. Computing $\vec{u} \times \vec{B}$ with the velocities above gives the signs and the extra $u'_z B_x/\gamma$ term written here (they agree with Experiment 1 and with $E'_z = \gamma(E_z + vB_y)$). The final $B'_y$ below, which uses only $F_x$, $F_z$ and $F'_x$, is the same as on the board.

### Using the force transformation: the $x$-component gives $B'_y$

Here $\vec{F} \cdot \vec{u} = F_x v + F_z \dfrac{u'_z}{\gamma}$ (since $u_y = 0$), and with $u_x = v$ as before

$$F'_x = \frac{F_x - \dfrac{v}{c^2}\left(F_x v + F_z \dfrac{u'_z}{\gamma}\right)}{1 - \dfrac{v^2}{c^2}} = F_x - \frac{\gamma\, v\, u'_z}{c^2} F_z$$

Substitute $F_x$ and $F_z$:

$$\begin{aligned} F'_x &= q\left[E_x - \frac{u'_z}{\gamma} B_y - \frac{\gamma\, v\, u'_z}{c^2}\left(E_z + vB_y\right)\right] \\ &= q\left[E_x - u'_z B_y \left(\frac{1}{\gamma} + \frac{\gamma v^2}{c^2}\right) - \frac{\gamma\, v\, u'_z}{c^2} E_z\right] = q\left[E_x - \gamma\, u'_z \left(B_y + \frac{v\, E_z}{c^2}\right)\right] \end{aligned}$$

Compare with $F'_x = q\left(E'_x - u'_z B'_y\right)$ and $E'_x = E_x$:

$$\boxed{B'_y = \gamma\left(B_y + \frac{v\, E_z}{c^2}\right)}$$

**where:** $B_y, E_z$ are measured in $S$; $B'_y$ in $S'$; $v$ is the velocity of $S'$ relative to $S$; $c$ is the speed of light; $\gamma = 1/\sqrt{1 - v^2/c^2}$.

**Consistency:** the $y$-component, $F'_y = \gamma F_y$, gives $q(E'_y + u'_z B'_x) = q\left[\gamma(E_y - vB_z) + u'_z B_x\right]$, i.e. $B'_x = B_x$ again, as found in Experiment 1.

## 4 Result and Generalisation

**Transformation of $\vec{B}$ for a boost along $x$**

$$B'_x = B_x, \qquad B'_y = \gamma\left(B_y + \frac{v\, E_z}{c^2}\right), \qquad B'_z = \gamma\left(B_z - \frac{v\, E_y}{c^2}\right)$$

**Generalising**, with $\parallel$ and $\perp$ taken with respect to the direction of $\vec{v}$ (the velocity of $S'$ relative to $S$):

$$\boxed{B'_\parallel = B_\parallel}, \qquad \boxed{\vec{B}'_\perp = \gamma\left(\vec{B}_\perp - \frac{1}{c^2}\left(\vec{v} \times \vec{E}\right)_\perp\right)}$$

**where:** $\vec{B}_\parallel$ is the component of $\vec{B}$ along $\vec{v}$, $\vec{B}_\perp$ the component perpendicular to $\vec{v}$; $\vec{v} \times \vec{E}$ is automatically perpendicular to $\vec{v}$.

**Check:** for $\vec{v} = v\hat{x}$, $\vec{v}\times\vec{E} = (0,\, -vE_z,\, vE_y)$, so $B'_y = \gamma(B_y + vE_z/c^2)$ and $B'_z = \gamma(B_z - vE_y/c^2)$, as derived.

**Complete set, for a boost along $x$ (with the previous lecture)**

$$E'_x = E_x, \quad E'_y = \gamma(E_y - vB_z), \qquad E'_z = \gamma(E_z + vB_y)$$
$$B'_x = B_x, \quad B'_y = \gamma\left(B_y + \frac{v}{c^2}E_z\right), \quad B'_z = \gamma\left(B_z - \frac{v}{c^2}E_y\right)$$

## 5 Check Using the Inverse Equations

The board remarks that the same results are obtained with the inverse equations. For Experiment 1 ($u'_x = 0$) the inverse force transformation (swap primed and unprimed, $v \to -v$) gives

$$F_x = F'_x + \frac{v}{c^2}\left(\vec{F}\,'\cdot\vec{u}\,'\right) = F'_x + \frac{v}{c^2}\,u'_y F'_y = q\left[E'_x + u'_y\left(B'_z + \frac{v}{c^2}E'_y\right)\right]$$

(using $F'_x = q(E'_x + u'_y B'_z)$ and $F'_y = qE'_y$). Equating to $F_x = q\left(E_x + \dfrac{u'_y}{\gamma}B_z\right)$ and using $E'_x = E_x$, $E'_y = \gamma(E_y - vB_z)$:

$$B'_z + \frac{\gamma v}{c^2}\,(E_y - vB_z) = \frac{B_z}{\gamma} \quad\Longrightarrow\quad B'_z = B_z\left(\frac{1}{\gamma} + \frac{\gamma v^2}{c^2}\right) - \frac{\gamma v}{c^2}E_y = \gamma\left(B_z - \frac{v\,E_y}{c^2}\right)$$

which is the same $B'_z$ as before.